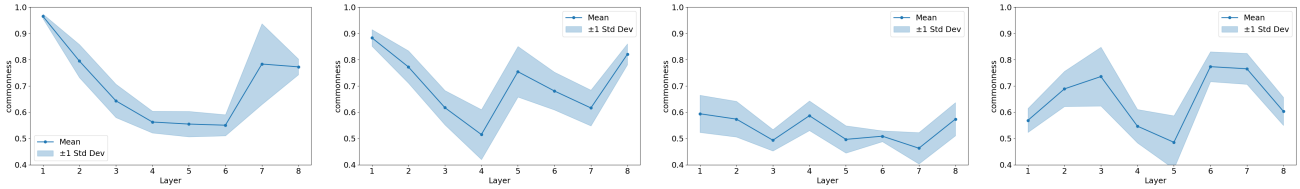
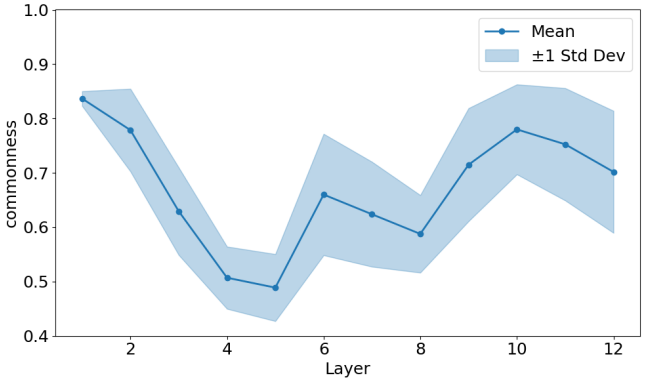
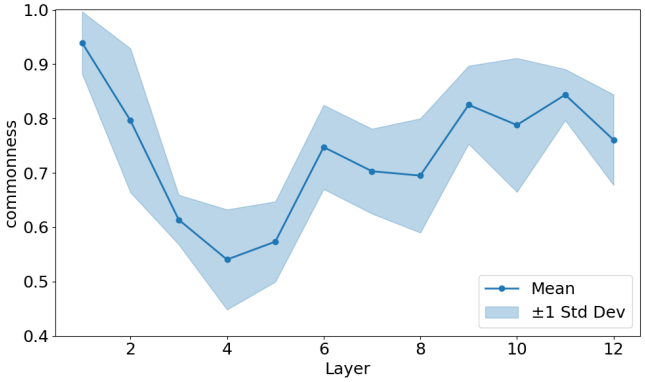


(a) Layers: 8, Heads: 8, Attn-only: False, Adam *(b)* Layers: 8, Heads: 8, Attn-only: False, AdamW *(c)* Layers: 8, Heads: 8, Attn-only: True, Adam *(d)* Layers: 8, Heads: 8, Attn-only: True, AdamW



(e) Layers: 12, Heads: 12, Attn-only: False, Adam

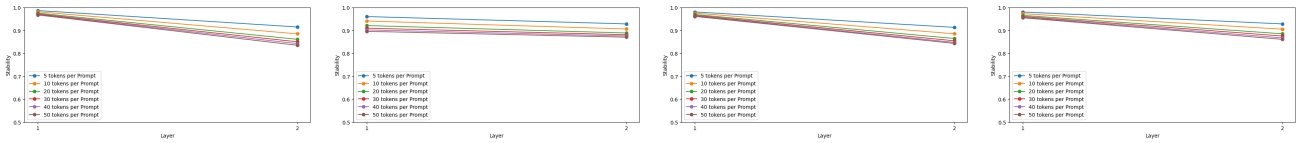


(f) Layers: 12, Heads: 12, Attn-only: False, AdamW

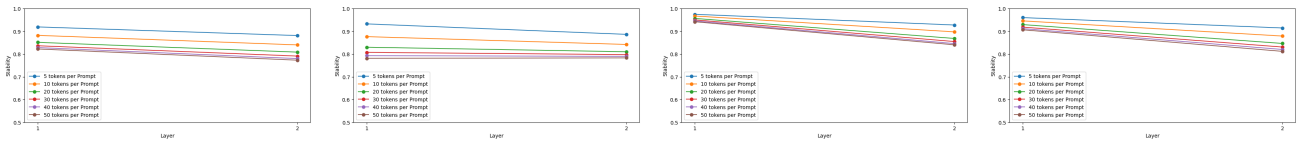
Figure 22. Representative plots for cross-layer best-match stability § 4.3

B.4. EFFECT OF PROMPT LENGTH ON STABILITY

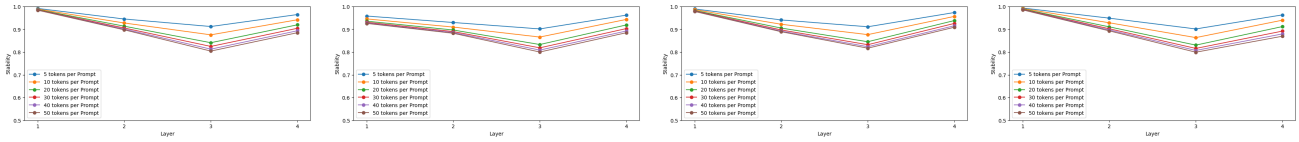
Additional architecture-wise plots for § 4.4.2:



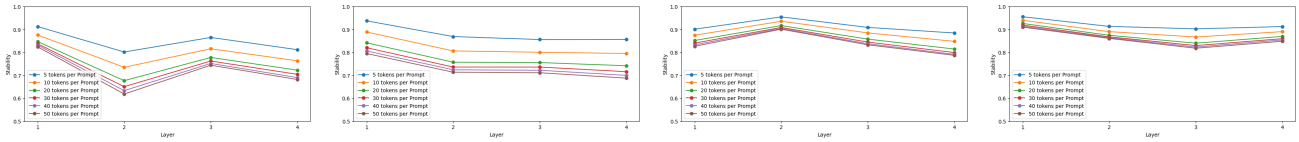
(a) Layers: 2, Heads: 8, Attn-only: False, Adam *(b)* Layers: 2, Heads: 16, Attn-only: False, Adam *(c)* Layers: 2, Heads: 8, Attn-only: False, AdamW *(d)* Layers: 2, Heads: 16, Attn-only: False, AdamW



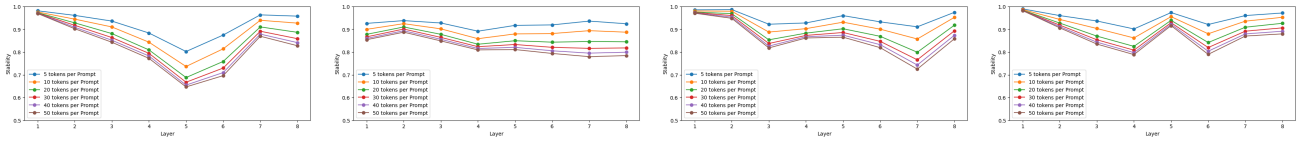
(a) Layers: 2, Heads: 8, Attn-only: True, Adam *(b)* Layers: 2, Heads: 16, Attn-only: True, Adam *(c)* Layers: 2, Heads: 8, Attn-only: True, AdamW *(d)* Layers: 2, Heads: 16, Attn-only: True, AdamW



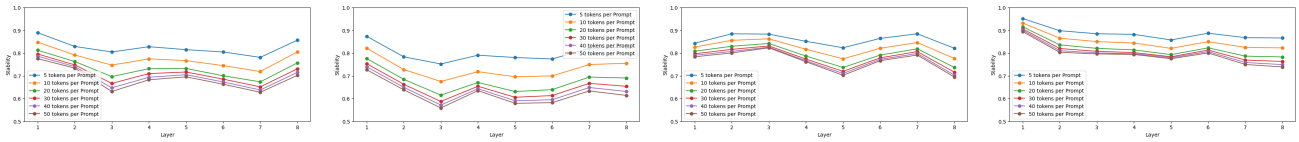
(a) Layers: 4, Heads: 8, Attn-only: False, Adam *(b)* Layers: 4, Heads: 16, Attn-only: False, Adam *(c)* Layers: 4, Heads: 8, Attn-only: False, AdamW *(d)* Layers: 4, Heads: 16, Attn-only: False, AdamW



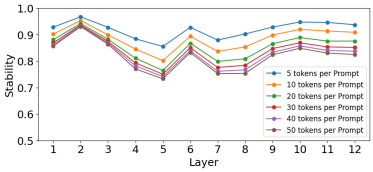
(a) Layers: 4, Heads: 8, Attn-only: True, Adam *(b)* Layers: 4, Heads: 16, Attn-only: True, Adam *(c)* Layers: 4, Heads: 8, Attn-only: True, AdamW *(d)* Layers: 4, Heads: 16, Attn-only: True, AdamW



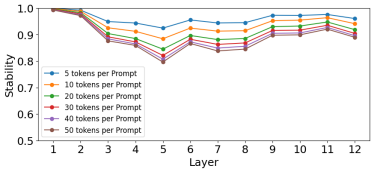
(a) Layers: 8, Heads: 8, Attn-only: False, Adam *(b)* Layers: 8, Heads: 16, Attn-only: False, Adam *(c)* Layers: 8, Heads: 8, Attn-only: False, AdamW *(d)* Layers: 8, Heads: 16, Attn-only: False, AdamW



(a) Layers: 8, Heads: 8, Attn-only: True, Adam *(b)* Layers: 8, Heads: 16, Attn-only: True, Adam *(c)* Layers: 8, Heads: 8, Attn-only: True, AdamW *(d)* Layers: 8, Heads: 16, Attn-only: True, AdamW



(a) Layers: 12, Heads: 12, Attn-only: False, Adam



(b) Layers: 12, Heads: 12, Attn-only: False, AdamW

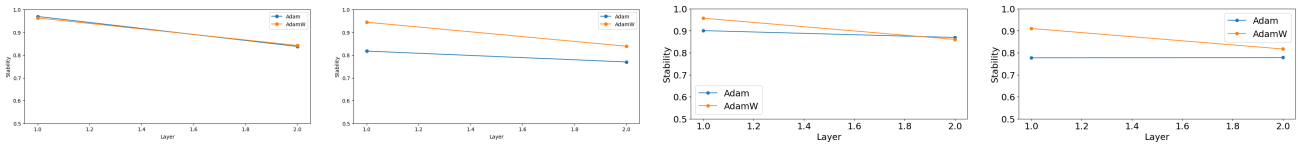
B.5. Correlation between Query-weight norm and layer-wise stability

Layer	Heads	Attention-only	Optimizer	Correlation
2	8	False	False	-1.0
2	8	False	True	-1.0
2	8	True	False	-1.0
2	8	True	True	-1.0
2	16	False	False	-1.0
2	16	False	True	-1.0
2	16	True	False	1.0
2	16	True	True	-1.0
4	8	False	False	-0.9406589554922179
4	8	False	True	-0.9015930143944082
4	8	True	False	-0.6827415649248102
4	8	True	True	-0.2212791321322467
4	16	False	False	-0.8397480100370013
4	16	False	True	-0.9434888997439926
4	16	True	False	-0.6494382579338498
4	16	True	True	-0.9554985039029008
8	8	False	False	-0.8447143648189646
8	8	False	True	-0.6931182527732535
8	8	True	False	-0.7659110184144368
8	8	True	True	-0.031545835341075135
8	16	False	False	-0.21617605236079285
8	16	False	True	-0.6749555081511857
8	16	True	False	-0.6941387312838879
8	16	True	True	-0.6374924846114093
12	12	False	False	-0.5078991103274401
12	12	False	True	-0.8228023018292414

Figure 30. Layer-wise stability is inversely correlated with average (ℓ_2) norm of attention-head query weights, suggesting a link between norm growth and head instability.

B.6. Stability comparison: Adam vs AdamW

Additional architecture-wise plots showing improvement in stability when AdamW optimizer is used while pre-training (§ 4.5):



(a) Layers: 2, Heads: 8, Attn-only: False *(b)* Layers: 2, Heads: 8, Attn-only: True *(c)* Layers: 2, Heads: 16, Attn-only: False *(d)* Layers: 2, Heads: 16, Attn-only: True